

reflection at 14 Å periodicity that is unaltered following glycolation. The reflection slightly collapsed after 550 °C heating shows the presence of vermiculite alongside with the other minerals. Quartz was identified based on reflection related to 3 Å periodicity in the data.

EDX

The quantitative EDX analysis (Table 1) suggested O and Si as dominant elements in the sample. These elements are common in both quartz and clay minerals. Al and K, which are common in illite and kaolinite are found in smaller quantities. Fe also found in small quantities, which confirms the presence of illite as indicated by XRD and FTIR data. Minor concentrations of Na and Mg were also found in the sample. Mg is common in illite but Na is usually found in smectite. Since the XRD data did not indicate the presence of smectite, Na found in the EDX data could be attributed to residual impurities.

FTIR

Using the benchtop FTIR the mineral spectrum of the samples was automatically compared to a spectral library stored in the equipment. The mineral content in the samples was produced as a percentage as in Fig. 2. Consistent with the XRD result, FTIR data also recorded the presence of quartz, illite, kaolinite and chlorite. For the purpose of analysis, 10% tolerance is accepted for comparison of mineral content in each sample. Samples from toluene and chloroform recorded the same amount of quartz, illite, kaolinite and chlorite (within the acceptable tolerance). Samples from methanol and azeotrope recorded more quartz and less illite. Kaolinite

and chlorite also presence in a smaller amount. The difference could be due to inefficient cleaning by methanol as inferred by the effluent's color. It had been observed that the effluent from samples in methanol was clear while effluent from toluene and chloroform were blackish. Effluent from azeotrope showed lighter color as compared to toluene and chloroform. This indicates that toluene and chloroform provide efficient cleaning while methanol is not suitable for this type of mud. This is supported by a study by Gupta et al. [2], who reported that toluene and chloroform have similar cleaning efficiencies. On the other hand, methanol is more suitable for removing polar components and dissipated salt [1]. The presence of mud contamination could have concealed the actual spectrum. The GCMS result indicates an abundance of alkane hydrocarbon (tridecane, undecane, ethyl, cyclopentane) and acid (oxalic and sulfuric) and no traces of components from

Table1. EDX result

No	Element	Weight %	Atomic %
1	O	56.53 ± 1.54	70.52
2	Na	0.24 ± 0.12	0.21
3	Mg	0.29 ± 0.12	0.24
4	Al	5.44 ± 0.25	4.02
5	Si	32.52 ± 0.95	23.11
6	K	1.39 ± 0.17	0.71
7	Ca	0.36 ± 0.14	0.18
8	Ti	0.27 ± 0.16	0.11
9	Fe	2.37 ± 0.38	0.85
10	U	0.58 ± 0.5	0.05
Total		100	100

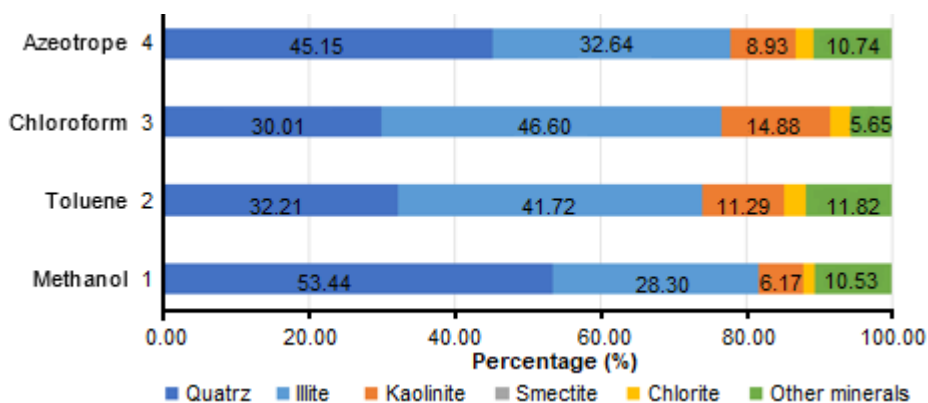


Fig 2. FTIR analysis result on percentage of clay minerals